

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Bioscience & Biotechnology

2. Subject Name: Microbiology L-T-P: 3-1-0 Credits: 4

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

1. Introduction to Microbiology, Basic concepts; Microbiological discoveries, Techniques and Applications; Scopes and relevance of Microbiology (03 lectures)
2. Microbial nutrition: (02 Lectures)
 - a. Common nutrient requirements; Requirements of carbon, hydrogen, oxygen and electrons
 - b. Nutritional types of microorganisms
 - c. Nutrient uptake by microorganisms
3. Microbial cell structure: (05 lectures)
 - a. Morphology and cell structure of prokaryotes and eukaryotes (bacteria; fungi, algae and viruses), Basic difference between a microbial cell and Eukaryotic cell
 - b. Structural organization of bacterial cell wall including peptidoglycan and lipopolysaccharides
 - c. Nuclear material and other cellular organelles
 - d. Bacterial biofilm
4. Microbial metabolism: (05 lectures)
 - a. Concepts of catabolism and anabolism; Chemoorganotrophic fueling reactions
 - b. Glycolysis; Fermentation; TCA cycles and Anaerobic respiration
 - c. Electron transport chain and oxidative phosphorylation
 - d. Chemolithotrophy and phototrophy
5. Microbial Growth: (04 lectures)
 - a. Concept of synchronous and non-synchronous growth
 - b. Bacterial cell division (Binary fission and the genes involved)
 - c. Sporulation and cell differentiation
 - d. Method of visualizing growth and use of microscopy to understand intricate cellular changes
 - e. Bacterial growth curve and the concepts underneath
 - f. Physical and organic factors influencing growth
 - g. Various method of sterilization

6. Growth inhibitory substances: (06 lectures)

- a. Concept of abiosis and various method; difference between disinfectant and antibiotic
- b. Types of chemical disinfectants and their action
- c. Different types of antimicrobials and their mechanisms of actions;
- d. Mechanism of development of antimicrobial resistance

7. Microbial Genetics and gene transfer: (03 lectures)

- a. Transformation, conjugation and transduction;
- b. Plasmids as vectors,
- c. Mutation and its types, deletions, base pair substitutions and frame shift mutations.

8. Microbial ecology, diversity and taxonomy (03 lectures)

- a. Microbial ecology and phylogeny
- b. Microbial classification and taxonomy
- c. Molecular approaches of microbial identification and classification

9. Viruses: (05 lecture)

- a. Introduction to viruses.
- b. Virus classification
- c. Structure and functional component of virus.
- d. viral replication
- e. Bacteriophage – replication and gene expression

10. Industrial microbiology; Applications in energy, environment, food & agriculture and health care (03 lectures)

Reference Books:

1. Prescott, Harley, and Klein's Microbiology by Joanne M. Willey, Linda M. Sherwood, Christopher J. Woolverton, Mc Graw Hill Higher Education, 7th Edition, 2007; ISBN 978-0073302089
2. Brock Biology of Microorganisms by Michael T Madigan, John M Martinko, Kelly S Bender, Daniel H Buckley and David A Stahl, Pearson Education, Limited, 14th Edition, 2015; ISBN 978-0-321-89739-8
3. Microbiology: Principles and Explorations by Jacquelyn G. Black, Laura J. Black, Wiley Publications, 10th Edition, ISBN: 978-1-119-39011-4

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures &
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		tutorial
1.	Introduction to microbiology	3
2.	Microbial nutrition	2 + 1
3.	Microbial cell structure	5 + 1
4.	Microbial metabolism	4 + 1
5	Microbial growth	4 + 1
6	Growth inhibitory substances:	6 + 1
7	Microbial Genetics and gene transfer	3 + 1
8	Microbial diversity and taxonomy	3 + 1
9	Viruses	5 + 1
10	Industrial microbiology; Applications in energy, environment, food & agriculture and health care	3 + 1
Total number of hours		38 + 9 = 47